<u>PROJECT 03 REPORT</u>

Automatic Street Light System using LDR

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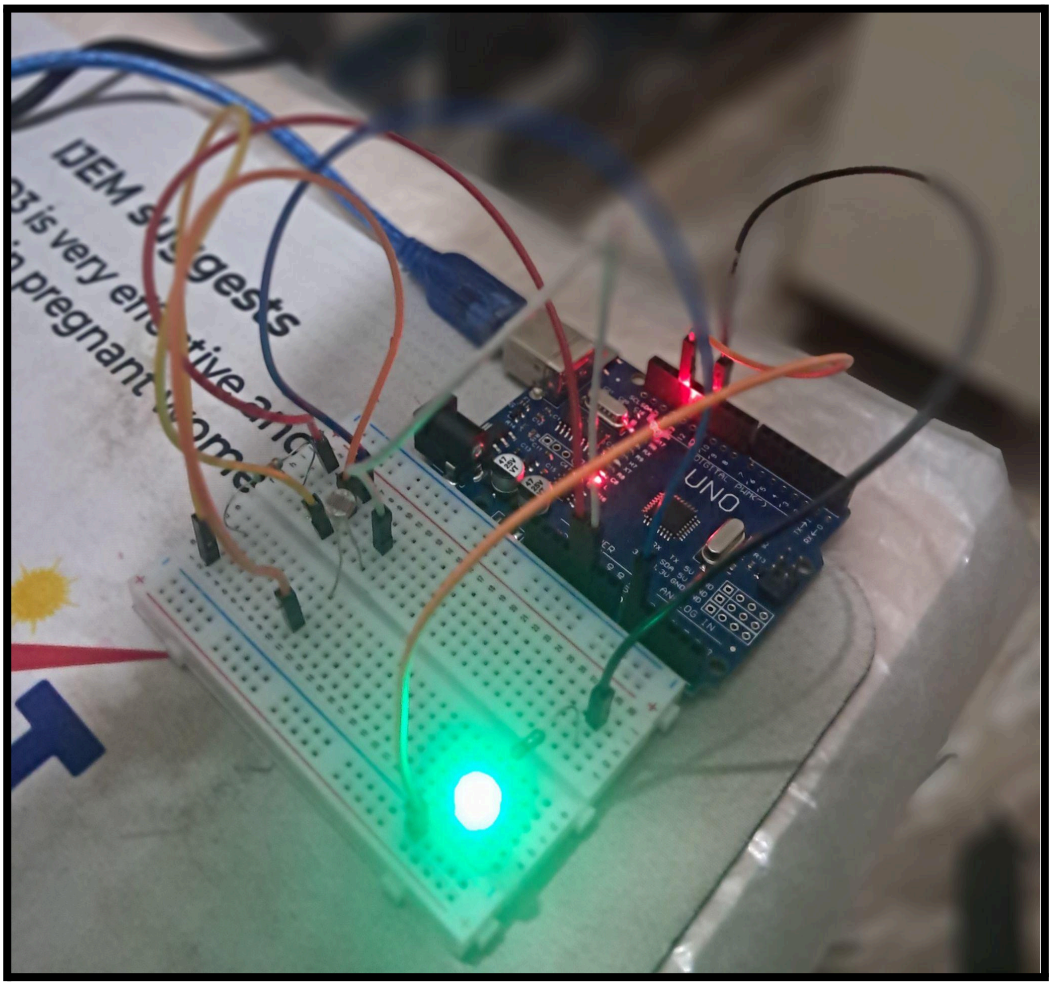
Duration : 2 days



1. Objective

The objective of this project was to develop an automatic street light system using an LDR (Light Dependent Resistor) and Arduino Uno. The system automatically detects surrounding light intensity and controls the LED brightness accordingly without any human interaction.

The project also introduced analog sensors, voltage dividers, ADC conversion, threshold-based decision making, and PWM - based brightness control.

2. Components Used

Component	Quantity
Arduino Uno R3	1
Breadboard	1
LDR (Light Dependent Resistor)	1
LEDs (Blue, Green)	1
Resistors(10kΩ , 300Ω)	1
Jumper Wires	Multiple
USB Cable	1

3. Software Used

- Arduino IDE
- Serial Monitor
- Windows Laptop

4. Theory

An LDR (Light Dependent Resistor) is a sensor whose resistance changes according to the intensity of light falling on its surface.

- Bright Light → Low Resistance
- Darkness → High Resistance

Arduino cannot directly measure resistance so a voltage divider circuit is used to convert the changing resistance into a varying voltage.

The voltage at the junction point is connected to Analog Pin A0 and read using: **analogRead(A0);**

Arduino Uno contains a 10-bit Analog to Digital Converter (ADC) which converts input voltages into values ranging from:

0V → 0

5V → 1023

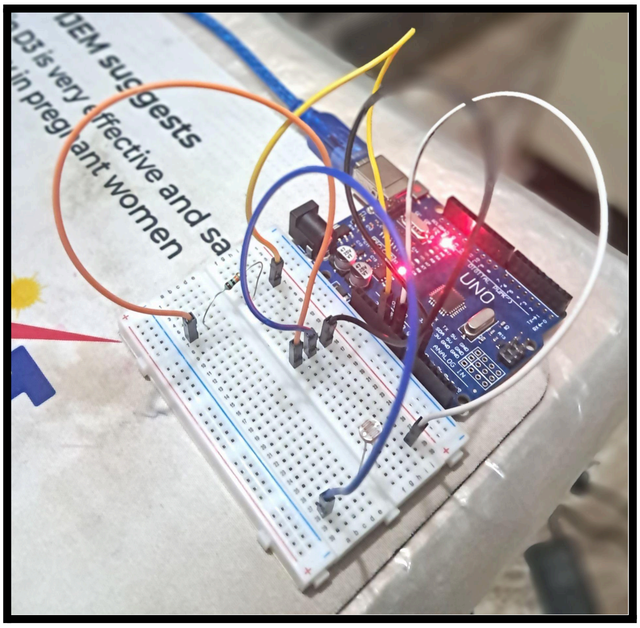
5. Tasks Performed

➢ Understanding LDR and analogRead()

The working principle of LDR, LDR resistance, voltage divider circuits, and analog to digital conversion was studied before hardware implementation.

The concept of **analogRead(A0)** and Arduino's 10-bit ADC was studied to understand how changing voltages are being converted into values ranging from 0 - 1023.

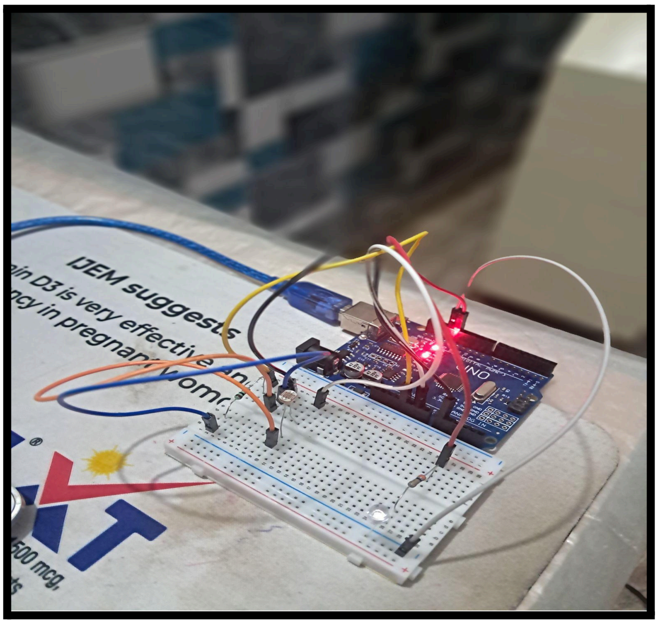
➢ Reading Live LDR Values



Video

The LDR was connected to Arduino using a voltage divider circuit and sensor readings were observed through the Serial Monitor under different lighting conditions .

➢ Automatic Street Light Logic

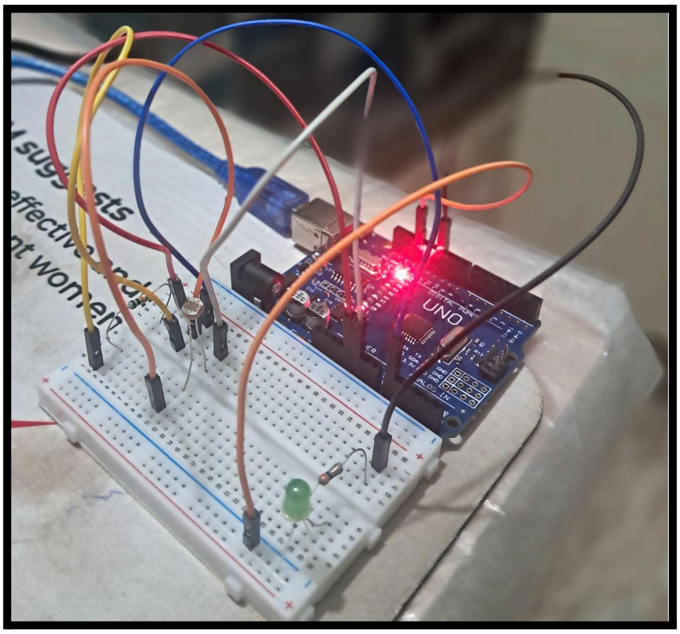


Video

An automatic street light system was implemented using the LDR and an LED. The Arduino continuously monitors the environment light intensity and compares the sensor value with a predefined threshold.

When the surroundings became dark, the LED automatically turned ON. When sufficient light was available, the LED remained OFF.

➢ PWM Based Brightness Control



Video

The automatic street light system was further improved using PWM (Pulse Width Modulation). Instead of simple ON/OFF control, three brightness levels were designed based on surrounding light conditions.

Different PWM duty cycles were used to vary LED brightness, creating a more realistic and efficient automatic lighting system.

6. Learning Outcomes

The following concepts were learned in making of this project:

- LDR Sensor Operation
- Voltage Divider Circuits
- Analog to Digital Conversion (ADC)

- analogRead()
- Threshold based Decision Making
- PWM Brightness Control
- Sensor based Automation
- Debugging Techniques
- Environmental Sensing
- Real-Time Embedded Control Systems

7. Problems Faced

I. Initially, understanding the voltage divider behavior and the relationship between LDR resistance, voltage, and analog values was challenging.

II. Selecting appropriate threshold values for automatic switching. **(Sensor readings were recorded under multiple lighting conditions for selecting threshold values experimentally)**

III. The LED responded slowly to environmental changes during testing. **(A debugging delay was added for observing Serial Monitor readings which was removed after calibration)**

8. Observations

- Bright light produced lower analog values.
- Darkness produced higher analog values.
- Sensor readings changed continuously with environmental lighting conditions.
- PWM enables multiple brightness levels instead of simple ON/OFF operation.

9. Applications and Future Scope

Automatic light sensing systems are widely used in street lights, garden lights, solar lighting systems, and smart energy saving solutions. These systems automatically adjust the lighting conditions without human interactions.

The project can be further enhanced by incorporating additional features such as ,Hysteresis based switching (buffer zone) to prevent flickering near threshold values, OLED/LCD display for real-time sensor monitoring and Solar powered autonomous lighting systems.

10. Conclusion

This project successfully demonstrated how an LDR sensor can be used to monitor surrounding light intensity and automatically control lighting conditions.

The project provides a strong foundation for future sensor interfacing and IoT projects .